ScanCrypt

Post-Quantum Cryptographic Security Assessment

Project Name	m4fstack
Analysis Date	2026-07-09 20:57:02
ScanCrypt Version	1.0
Algorithm	ML-KEM / Kyber
Security Level	Unknown ML-KEM Variant
Implementation Type	Masked Implementation

High

Confidentiality Notice

This document contains confidential assessment details based on ScanCrypt analysis. Unauthorized disclosure is prohibited.

Report generated automatically by ScanCrypt.

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Executive Summary

Overall Risk	High	**Algorithm**	ML-KEM / Kyber
Security Level	Unknown ML-KEM Variant	**Implementation Type**	Masked Implementation
Programming Language	C/C++	**Build System**	Not detected
Files	3	**Source Files**	2
Header Files	1	**Lines of Code**	906
Rules Executed	15	**Rules Passed**	9
Rules Failed	6	**Analysis Confidence**	99
Detection Confidence	96%	**Analysis Duration**	00:01:25

Critical: 0

High: 1

Medium: 0

Low: 0

Information: 0

Project Inspection

Algorithm	ML-KEM / Kyber
Security Level	Unknown ML-KEM Variant
Implementation Type	Masked Implementation
Programming Language	C/C++
Build System	Not detected
Files	3
Source Files	2
Header Files	1
Lines of Code	906
Detection Confidence	96%

Detection Evidence

- KYBER_N detected in poly.h
- KYBER_Q detected in poly.c
- KYBER_K detected in poly.c
- poly_compress() detected in poly.h
- poly_decompress() detected in poly.h
- ntt() detected in poly.h
- invntt() detected in poly.h

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Findings Summary

Rule ID	Severity	Category	Status	File	Line
IMP-001	High	Configuration	Failed	poly.c	42

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Detailed Findings

IMP-001 — High

KYBER_K parameter disclosure

Rule ID	IMP-001
Severity	High
Category	Configuration
Affected Algorithm	ML-KEM / Kyber
Affected File	poly.c
Affected Line	42

Description
KYBER_K parameter was detected but not validated against expected values.

Potential Security Impact
Incorrect parameter values can reduce algorithmic security.

Technical Explanation
Kyber security level depends directly on KYBER_K. The extracted KYBER_K value was missing or malformed.

Research-based Mitigation Recommendation
Ensure KYBER_K is defined with a supported integer value of 2, 3, or 4.

References
NIST PQC Kyber specification

RFC 9420

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Rule Execution Summary

✗ IMP-001 — Failed

Statistics

Total rules executed
15

Passed rules
9

Failed rules
6

Analysis duration
00:01:25

Files analysed
3

Source lines analysed
906

Execution Progress

Passed: 9 / 15



Failed: 6 / 15



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Appendix

Analysed Files

poly.c

poly.h

main.c

Tool version
1.0

Report generated
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Scan configuration
Default ScanCrypt profile

Supported algorithms
ML-DSA (Dilithium), ML-KEM (Kyber)

Supported implementation types
Portable C / Reference, AVX2 Optimized, ARM Cortex-M / Embedded, Masked Implementation

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